Signals and Systems
SET214
Lecture (02)- A Quick Tutorial on MATLAB

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Outlines

- Introduction
- MATLAB Interactive Sessions
- MATLAB Variable Names
- MATLAB Operators
- MATLAB Matrices
- Matrices Operation
- Graphics Fundamentals
- M-File Scripts
- Commands for Managing a Session
- Flow Control

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Introduction

- MATLAB is a software package for doing numerical computation.
- It was originally designed for solving linear algebra type problems using matrices.
- It's name is derived from MATrix LABoratory.
- MATLAB has since been expanded and now has built-in functions for solving problems requiring data analysis, signal processing, optimization, and several other types of scientific computations.
- It also contains functions for 2-D and 3-D graphics and animation.

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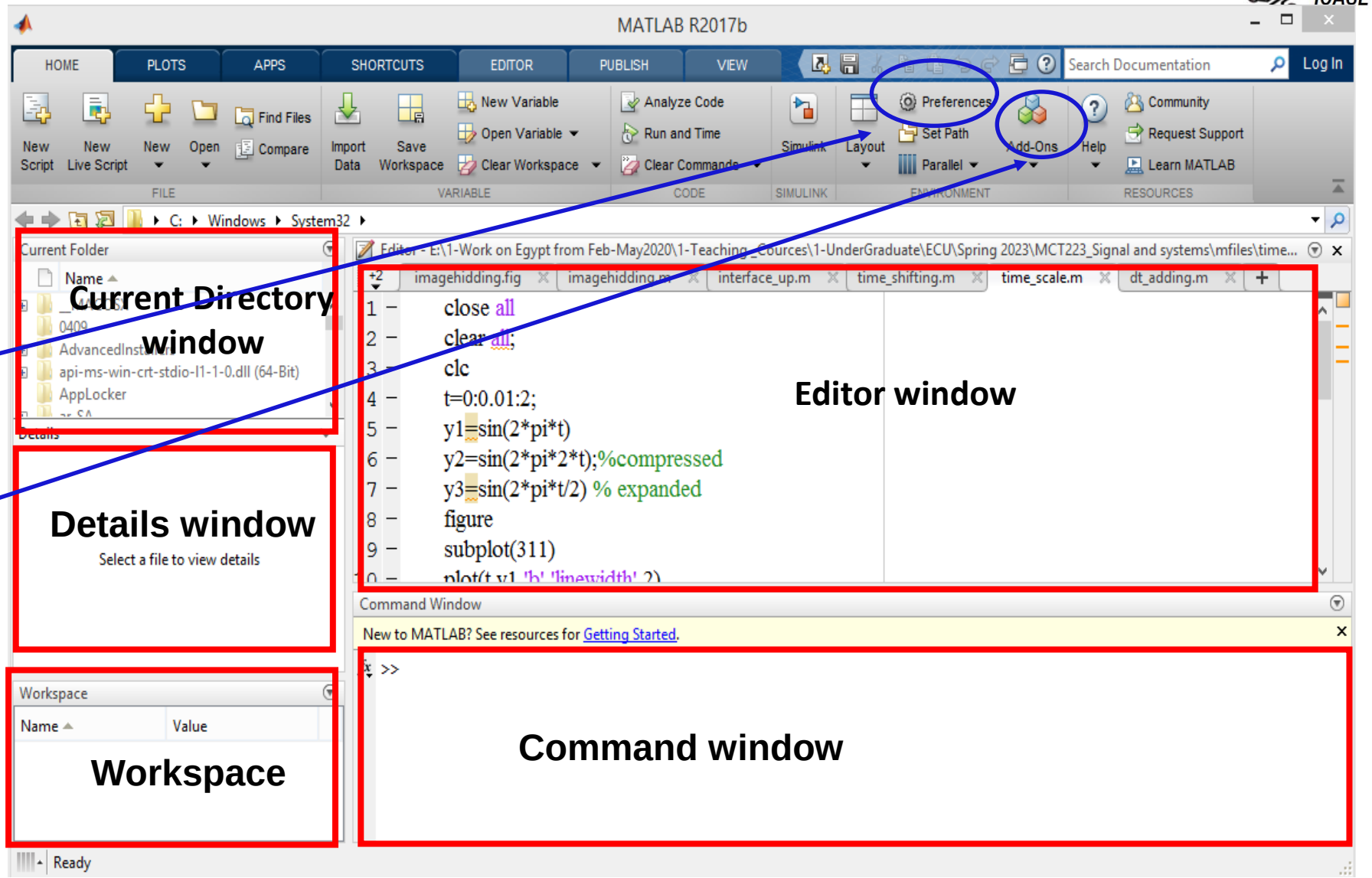
MATLAB Interactive Sessions

- **Five windows appear.**

1. Command window
2. Workspace
3. Details window
4. Current Directory window
5. Editor window

- Allows you to customize yout MATLAB experience (color, fonts, ..etc)
- Allows you to install toolboxes included.
- Recommended tool box:

- Curve fitting
- Computer vision system
- Optimization
- Signal processing



Default MATLAB Desktop

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MATLAB Variable Names

- MATLAB is a software package for doing numerical computation.
- Variable names are case sensitive.
- Variable names can contain up to 63 characters
- Variable names must start with a letter and can be followed by letters, digits and underscores.

Ex.,

```
>> x = 2;
>> abc_123 = 0.005;
>> 1ab = 2;
Error: Unexpected MATLAB expression
```

Special variables and constants

Command	Description
ans	Temporary variable containing the most recent answer.
eps	Speci es the accuracy of oating point precision.
i,j	The imaginary unit $\sqrt{-1}$.
Inf	In nity .
NaN	Indicates an unde ned numerical result.
pi	The number π.

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MATLAB Operators (1/2)

❑ Relational operators

Relational operator	Meaning
<	Less than.
<=	Less than or equal to.
>	Greater than.
>=	Greater than or equal to.
==	Equal to.
~=	Not equal to. (NOT != like in C)

❑ MATLAB supports three logical operators

Operator	Name	De nition
~	NOT	~A returns an array of the same dimension as A; the new array has 1s where A is 0 and 0s where A is nonzero.
&	AND	A & B returns an array of the same dimension as A and B; the new array has 1s where both A and B have nonzero elements and 0s where either A or B is 0.
|	OR	A | B returns an array of the same dimension as A and B; the new array has 1s where at least one element in A or B is nonzero and 0s where A and B are both 0.
&&	Short-Circuit AND	Operator for scalar logical expressions. A && B returns true if both A and B evaluate to true, and false if they do not.
||	Short-Circuit OR	Operator for scalar logical expressions. A || B returns true if either A or B or both evaluate to true, and false if they do not.

MATLAB Operators (2/2)

❑ Scalar arithmetic operations

Symbol	Operation	MATLAB form
^	exponentiation: a^b	a^b
*	multiplication: ab	a*b
/	right division: $a/b = \frac{a}{b}$	a/b
\	left division: $a\backslash b = \frac{b}{a}$	a\b
+	addition: $a + b$	a+b
−	subtraction: $a - b$	a−b

❑ Order of precedence

Precedence	Operation
First	Parentheses, evaluated starting with the innermost pair.
Second	Exponentiation, evaluated from left to right.
Third	Multiplication and division with equal precedence, evaluated from left to right.
Fourth	Addition and subtraction with equal precedence, evaluated from left to right.

To avoid mistakes, feel free to insert parentheses wherever you are unsure of the effect precedence will have on the calculation

```
>>8 + 3*5
ans =
     23
>>(8 + 3)*5
ans =
     55
>>4^2 - 12 - 8/4*2
ans =
      0
>>4^2 - 12 - 8/(4*2)
ans =
      3
>>3*4^2 + 5
ans =
     53
>>(3*4)^2 + 5
ans =
    149
>>27^(1/3) + 32^(0.2)
ans =
      5
>>27^(1/3) + 32^0.2
ans =
      5
>>27^1/3 + 32^0.2
ans =
     11
```

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MATLAB Matrices (1/4)

- MATLAB treats all variables as matrices.
- For our purposes a matrix can be thought of as an array, in fact, that is how it is stored.
- Vectors are special forms of matrices and contain only one row OR one column.

- Scalars are matrices with only one row AND one column.
  - A scalar can be created in MATLAB as follows:



MATLAB Matrices (2/4)

- A matrix with only one column is called a column vector. A column vector can be created in MATLAB as follows:



```
>> z = [12;10;-3]
z =
    12
    10
    -3
```

- A matrix with only one row is called a row vector. A row vector can be created in MATLAB as follows (note the commas):



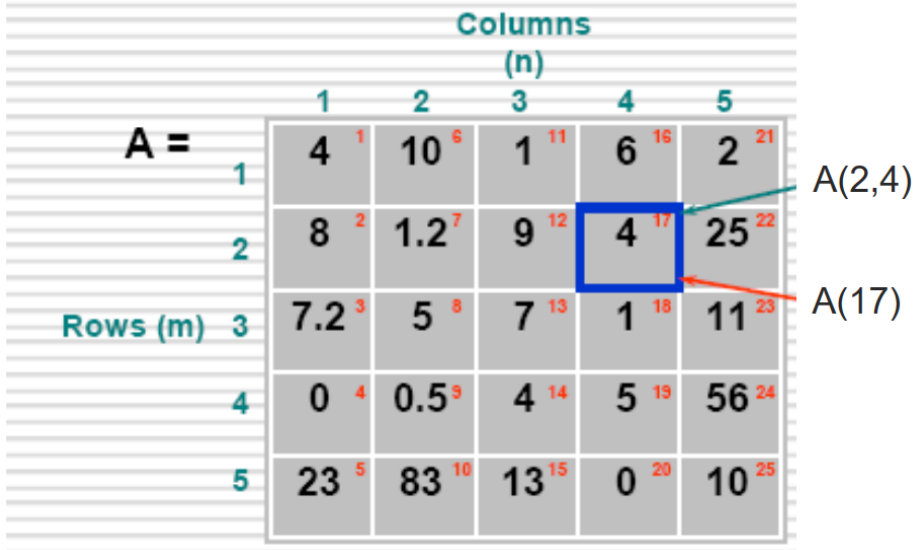
```
>> y = [12,10,-3]
y =
    12    10    -3
```

MATLAB Matrices (3/4)

- A matrix can be created in MATLAB as follows (note the commas and semicolons)

```
>> X = [1,2,3;4,5,6;7,8,9]
X =
     1     2     3
     4     5     6
     7     8     9
```

- Extracting a value from matrix.
  - A(2,4)=4
  - A(17)=4



MATLAB Matrices (4/4)

- A portion of a matrix can be extracted and stored in a smaller matrix by specifying the names of both matrices and the rows and columns to extract.
- The syntax is: sub_matrix = matrix (r1 : r2 , c1 : c2) ;
  - where r1 and r2 specify the beginning and ending rows and c1 and c2 specify the beginning and ending columns to be extracted to make the new matrix.

```
>> X = [1,2,3;4,5,6;7,8,9]
X =
     1     2     3
     4     5     6
     7     8     9
```

```
>> X13 = X(3,1:3)
X13 =
     7     8     9
```

```
>> X22 = X(1:2 , 2:3)
X22 =
     2     3
     5     6
```

```
>> X21 = X(1:2,1)
X21 =
     1
     4
```

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Matrices Operation (1/3)

❑ Matrix multiplication

```
>> a = [1,2;3,4];      (2x2)
>> b = [1,1];          (1x2)
>> c = b*a
c =
      4    6

>> c = a*b
??? Error using ==> mtimes
Inner matrix dimensions
must agree.
```

❑ Element wise multiplication

```
>> a = [1,2;3,4];
>> b = [1,½;1/3,¼];
>> c = a.*b
c =
    1    1
    1    1
```

Matrices Operation (2/3)

❑ Element wise division

```
>> a = [1,2;1,3];
>> b = [2,2;2,1];

>> c = a./b
c =
    0.5    1
    0.5    3
```

❑ Element wise power operation

```
>> c = a.^2
c =
    1    4
    1    9
```

Matrices Operation (3/3)

❑ Special matrices

Command	Description
`eye(n)`	Creates an $n \times n$ identity matrix.
`eye(size(A))`	Creates an identity matrix the same size as the matrix **A**.
`ones(n)`	Creates an $n \times n$ matrix of 1s.
`ones(m,n)`	Creates an $m \times n$ array of 1s.
`ones(size(A))`	Creates an array of 1s the same size as the array **A**.
`zeros(n)`	Creates an $n \times n$ matrix of 0s.
`zeros(m,n)`	Creates an $m \times n$ array of 0s.
`zeros(size(A))`	Creates an array of 0s the same size as the array **A**.

❑ Built in functions MATLAB

`cos(x)`	Cosine	`abs(x)`	Absolute value
`sin(x)`	Sine	`sign(x)`	Signum function
`tan(x)`	Tangent	`max(x)`	Maximum value
`acos(x)`	Arc cosine	`min(x)`	Minimum value
`asin(x)`	Arc sine	`ceil(x)`	Round towards $+\infty$
`atan(x)`	Arc tangent	`floor(x)`	Round towards $-\infty$
`exp(x)`	Exponential	`round(x)`	Round to nearest integer
`sqrt(x)`	Square root	`rem(x)`	Remainder after division
`log(x)`	Natural logarithm	`angle(x)`	Phase angle
`log10(x)`	Common logarithm	`conj(x)`	Complex conjugate



sort (X)
sum(X)
prod(X)
max(X)
min(X)
mean(X)
std(X)

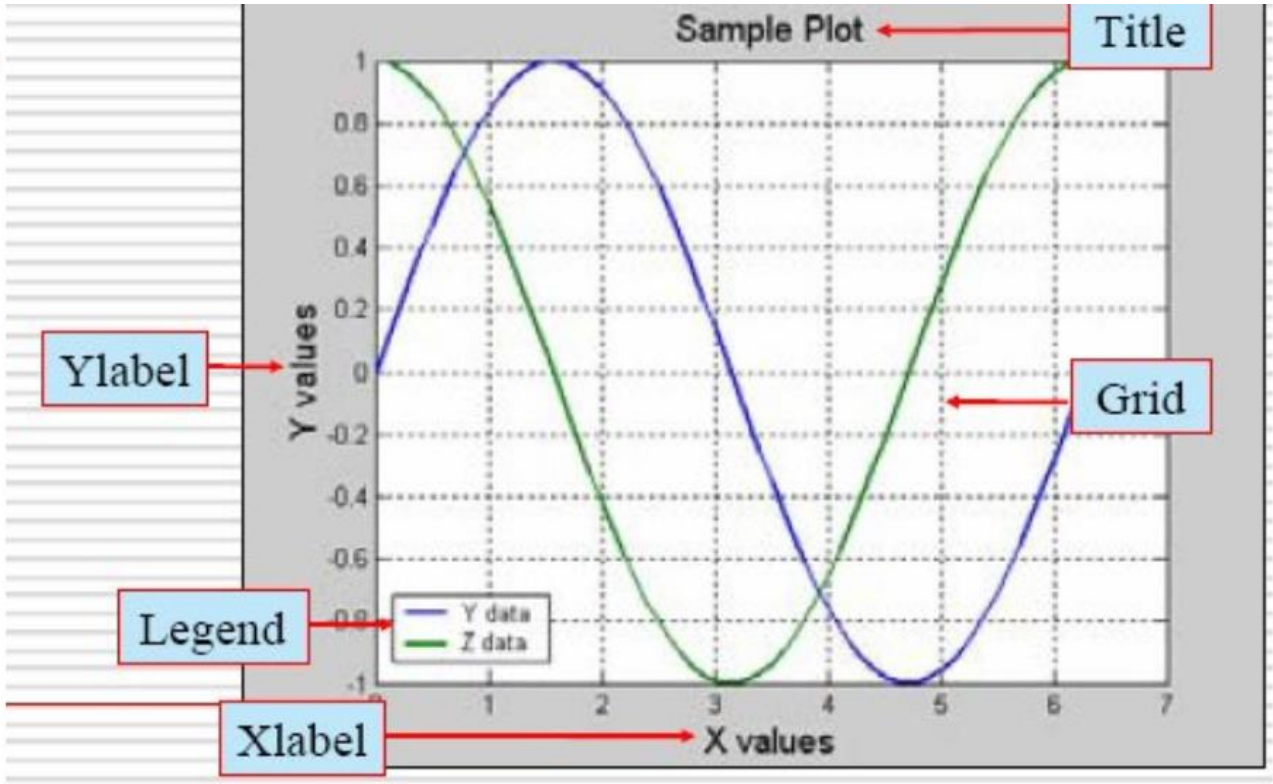
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Graphics Fundamentals (1/3)

- In 2-D plot, Plot sin(x) and cos(x) over [0,2π], on the same plot with different colors

```
>> x = linspace(0,2*pi,1000);
>> y = sin(x);
>> z = cos(x);
>> hold on;
>> plot(x,y,'b');
>> plot(x,z,'g');
>> xlabel 'X values';
>> ylabel 'Y values';
>> title 'Sample Plot';
>> legend ('Y data','Z data');
```



- Try to replace the 1st line by `>> x = 0:0.01:2*pi;`
- Check the result.

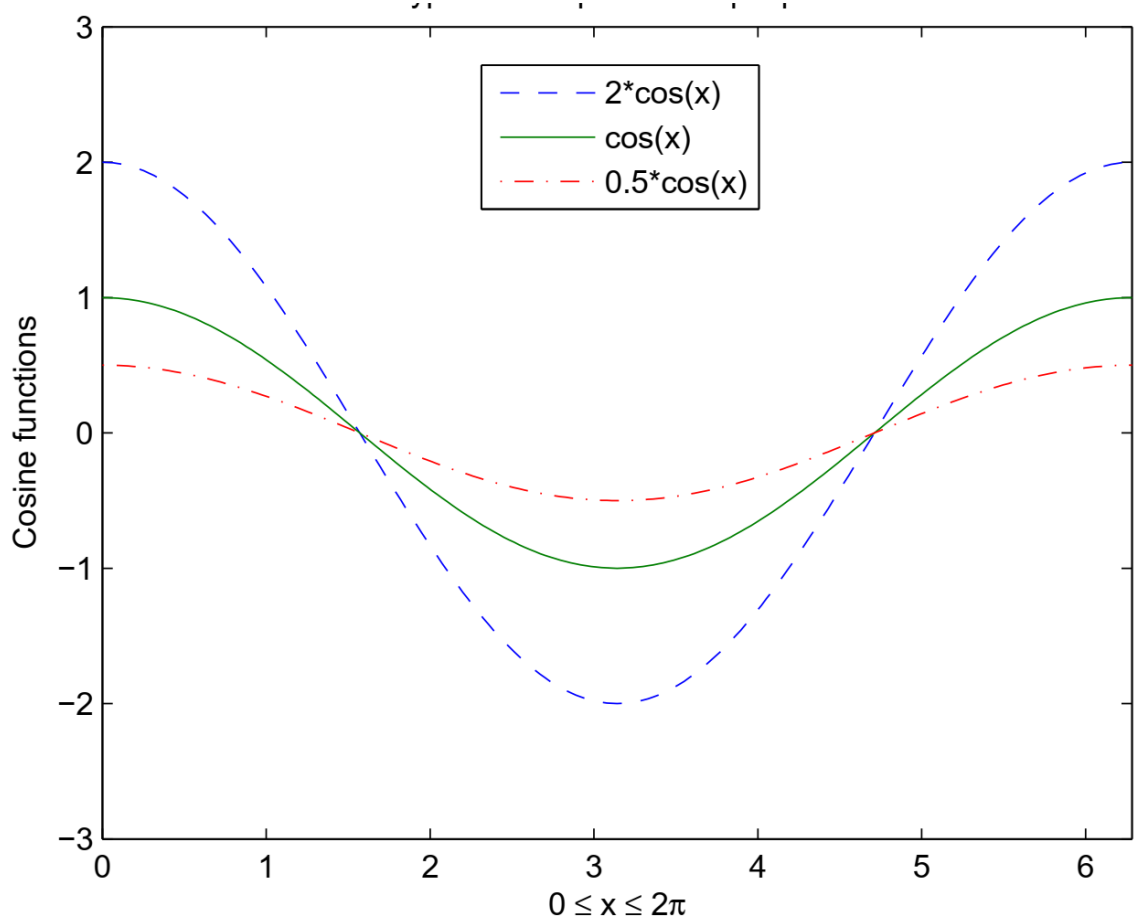
Please check in lab/section using Subplots function

Graphics Fundamentals (2/3)

- In 2-D plot, plot three related functions of x: y1 = 2 cos(x), y2 = cos(x), and y3 = 0:5 / cos(x), in the interval $0 \leq x \leq 2\pi$

```
>> x = 0:pi/100:2*pi;
>> y1 = 2*cos(x);
>> y2 = cos(x);
>> y3 = 0.5*cos(x);
>> plot(x,y1,'--',x,y2,'-',x,y3,':')
>> xlabel('0 \leq x \leq 2\pi')
>> ylabel('Cosine functions')
>> legend('2*cos(x)','cos(x)','0.5*cos(x)')

>> title('Typical example of multiple plots')
>> axis([0 2*pi -3 3])
```

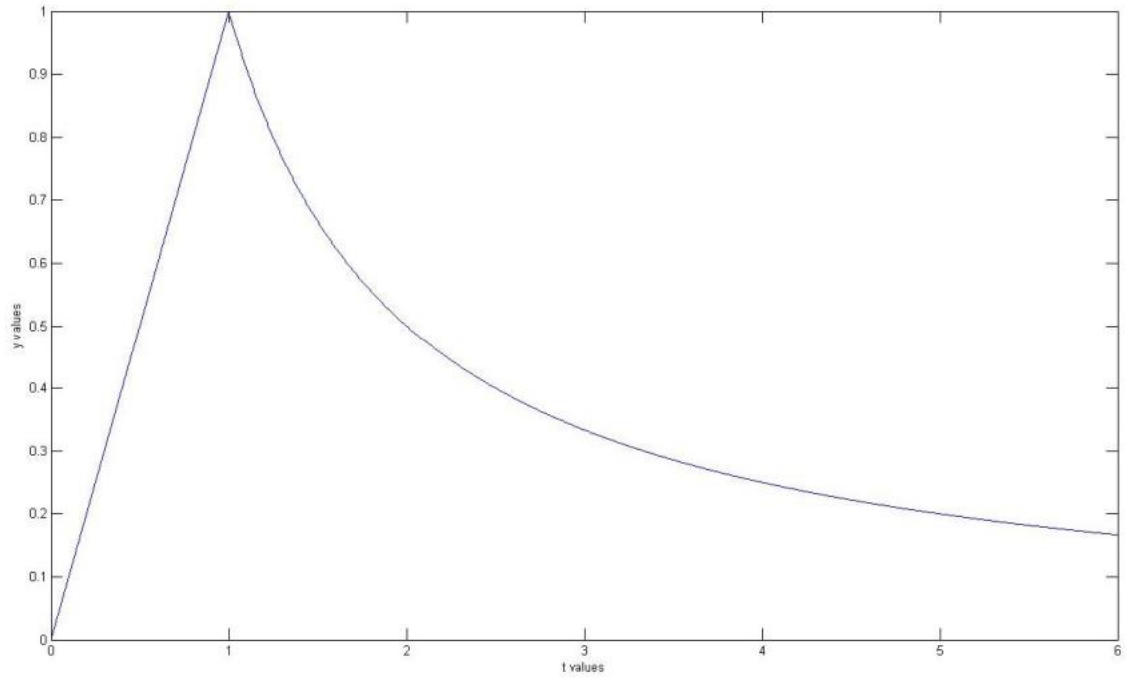


Graphics Fundamentals (3/3)

- In 2-D plot, Plot the following function:-

$$y = \begin{cases} t & 0 \le t \le 1 \\ 1/t & 1 \le t \le 6 \end{cases}$$

```
>> t1 = linspace(0,1,1000);
>> t2 = linspace(1,6,1000);
>> y1 = t1;
>> y2 = 1./ t2;
>> t = [t1,t2];
>> y = [y1,y2];
>> figure
>> plot(t,y);
>> xlabel 't values', ylabel 'y values';
```



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M-File Scripts (1/2)

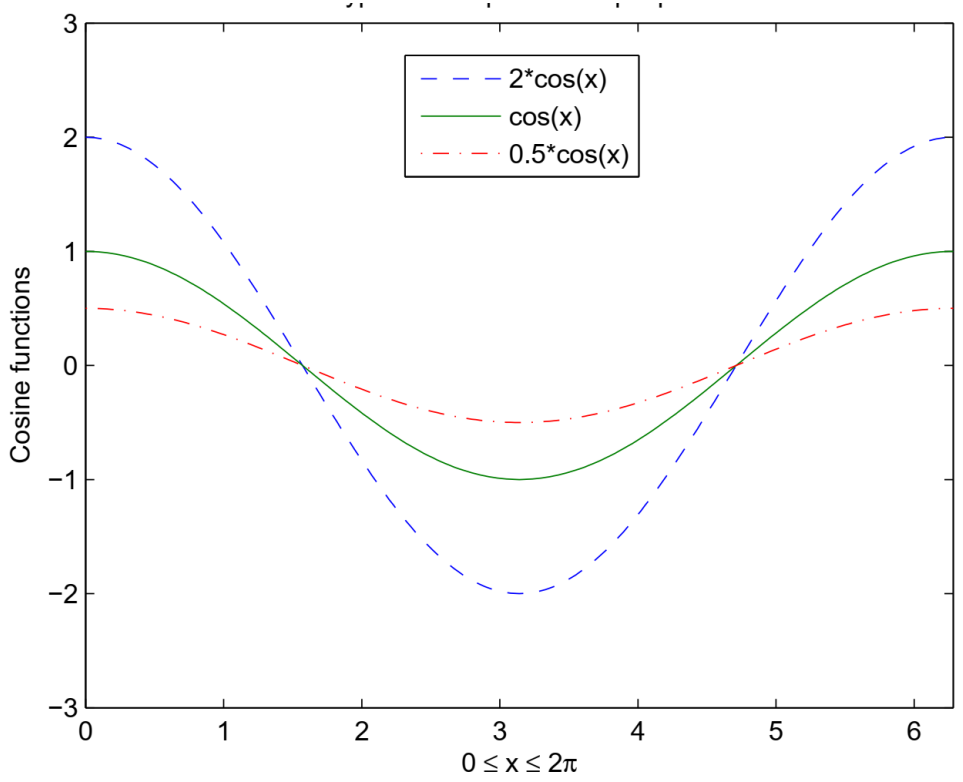
- Previously, all the commands were executed in the Command Window.
- A different way of executing repeatedly commands with MATLAB is :-
  1.to create a file with a list of commands,
  2. save the file, and
  3. run the file.
- If needed, corrections or changes can be made to the commands in the file. The files that are used for this purpose are called script files or scripts for short.
- M-File Scripts
  - A script file is an external flle that contains a sequence of MATLAB statements. Script files have a filename extension .m and are often called M-flles. M-flles can be scripts that simply execute a series of MATLAB statements, or they can be functions that can accept arguments and can produce one or more outputs.

M-File Scripts (2/1)

- In 2-D plot, plot three related functions of x: y1 = 2 cos(x), y2 = cos(x), and y3 = 0:5 / cos(x), in the interval $0 \leq x \leq 2\pi$. Using m-file.

- 1-Create a file , say example.m, which contains the following commands:

```
x = 0:pi/100:2*pi;
y1 = 2*cos(x);
y2 = cos(x);
y3 = 0.5*cos(x);
plot(x,y1,'--',x,y2,'-',x,y3,':')
xlabel('0 \leq x \leq 2\pi')
ylabel('Cosine functions')
legend('2*cos(x)','cos(x)','0.5*cos(x)')
title('Typical example of multiple plots')
axis([0 2*pi -3 3])
```



2-Run the file by typing example2 in the Command Window.

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Commands for Managing a Session

- MATLAB provides various commands for managing a session.
- The following table provides all such commands:

Command	Purpose
clc	Clears command window.
clear	Removes variables from memory.
exist	Checks for existence of file or variable.
global	Declares variables to be global.
help	Searches for a help topic.
lookfor	Searches help entries for a keyword.
quit	Stops MATLAB.
who	Lists current variables.
whos	Lists current variables (long display)

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Flow Control

- MATLAB has five flow control statements
  - if statements
  - switch statements
  - for loops
  - while loops
  - break statements

Will be covered in Lab/section

Thanks!

Summary

Table A.1: **Arithmetic operators and special characters**

Character	Description
+	Addition
−	Subtraction
*	Multiplication (scalar and array)
/	Division (right)
^	Power or exponentiation
:	Colon; creates vectors with equally spaced elements
;	Semi-colon; suppresses display; ends row in array
,	Comma; separates array subscripts
...	Continuation of lines
%	Percent; denotes a comment; specifies output format
′	Single quote; creates string; specifies matrix transpose
=	Assignment operator
( )	Parentheses; encloses elements of arrays and input arguments
[ ]	Brackets; encloses matrix elements and output arguments

Summary

Table A.2: **Array operators**

Character	Description
.*	Array multiplication
./	Array (right) division
.^	Array power
.\	Array (left) division
.′	Array (nonconjugated) transpose

Table A.3: **Relational and logical operators**

Character	Description
<	Less than
≤	Less than or equal to
>	Greater than
≥	Greater than or equal to
==	Equal to
~=	Not equal to
&	Logical or element-wise AND
|	Logical or element-wise OR
&&	Short-circuit AND
||	Short-circuit OR

Summary

Table A.4: Managing workspace and file commands

Command	Description
`cd`	Change current directory
`clc`	Clear the Command Window
`clear (all)`	Removes all variables from the workspace
`clear x`	Remove x from the workspace
`copyfile`	Copy file or directory
`delete`	Delete files
`dir`	Display directory listing
`exist`	Check if variables or functions are defined
`help`	Display help for MATLAB functions
`lookfor`	Search for specified word in all help entries
`mkdir`	Make new directory
`movefile`	Move file or directory
`pwd`	Identify current directory
`rmdir`	Remove directory
`type`	Display contents of file
`what`	List MATLAB files in current directory
`which`	Locate functions and files
`who`	Display variables currently in the workspace
`whos`	Display information on variables in the workspace

Summary

Table A.5: **Predefined variables and math constants**

Variable	Description
`ans`	Value of last variable (answer)
`eps`	Floating-point relative accuracy
`i`	Imaginary unit of a complex number
`Inf`	Infinity (∞)
`eps`	Floating-point relative accuracy
`j`	Imaginary unit of a complex number
`NaN`	Not a number
`pi`	The number π (3.14159...)

Table A.6: **Elementary matrices and arrays**

Command	Description
`eye`	Identity matrix
`linspace`	Generate linearly space vectors
`ones`	Create array of all ones
`rand`	Uniformly distributed random numbers and arrays
`zeros`	Create array of all zeros

Summary

Table A.7: Arrays and Matrices: Basic information

Command	Description
`disp`	Display text or array
`isempty`	Determine if input is empty matrix
`isequal`	Test arrays for equality
`length`	Length of vector
`ndims`	Number of dimensions
`numel`	Number of elements
`size`	Size of matrix

Summary

Table A.8: **Arrays and Matrices: operations and manipulation**

Command	Description
`cross`	Vector cross product
`diag`	Diagonal matrices and diagonals of matrix
`dot`	Vector dot product
`end`	Indicate last index of array
`find`	Find indices of nonzero elements
`kron`	Kronecker tensor product
`max`	Maximum value of array
`min`	Minimum value of array
`prod`	Product of array elements
`reshape`	Reshape array
`sort`	Sort array elements
`sum`	Sum of array elements
`size`	Size of matrix

Summary

Table A.9: Arrays and Matrices: matrix analysis and linear equations

Command	Description
`cond`	Condition number with respect to inversion
`det`	Determinant
`inv`	Matrix inverse
`linsolve`	Solve linear system of equations
`lu`	LU factorization
`norm`	Matrix or vector norm
`null`	Null space
`orth`	Orthogonalization
`rank`	Matrix rank
`rref`	Reduced row echelon form
`trace`	Sum of diagonal elements

Summary

Table A.9: Arrays and Matrices: matrix analysis and linear equations

Command	Description
`cond`	Condition number with respect to inversion
`det`	Determinant
`inv`	Matrix inverse
`linsolve`	Solve linear system of equations
`lu`	LU factorization
`norm`	Matrix or vector norm
`null`	Null space
`orth`	Orthogonalization
`rank`	Matrix rank
`rref`	Reduced row echelon form
`trace`	Sum of diagonal elements